



Comparing EKF and UT

Consider the measurement model,

$$\mathbf{y} = \mathbf{h}(\mathbf{x}) + \mathbf{r},$$

where $\mathbf{h}$ is assumed to be non-linear and $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$.

Objective

To find $\mathbb{E}(\mathbf{h}(\mathbf{x}))$ and $\text{Cov}(\mathbf{h}(\mathbf{x}))$, such that we can perform the moment matching

$$p(\mathbf{y}) \approx \mathcal{N}(\mathbf{y}; \mathbb{E}(\mathbf{h}(\mathbf{x})), \text{Cov}(\mathbf{h}(\mathbf{x})) + \mathbf{R}).$$

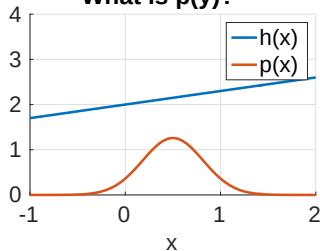
In this problem, we assume that x and y are scalar, and $r = 0$, such that $p(y)$ can be approximated by moment matching as:

$$p(y) \approx \mathcal{N}(y; \mathbb{E}(h(x)), \text{Var}(h(x)))$$

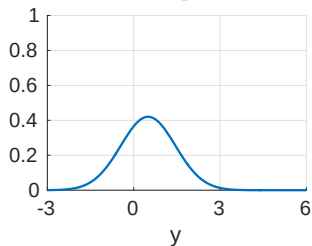
EXAMPLE 1 – WHAT IS $p(y)$ =?

CHALMERS

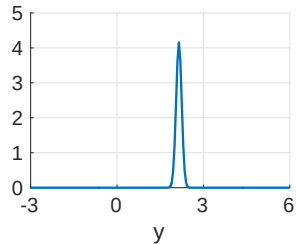
What is $p(y)$?



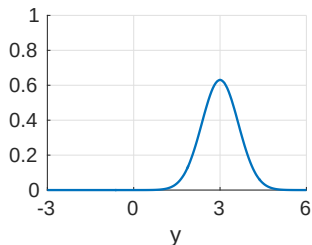
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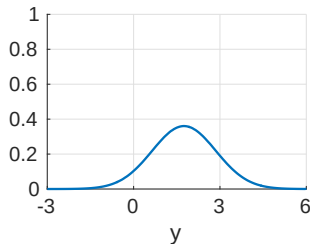
Yellow



Pink

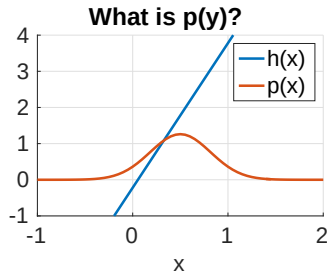


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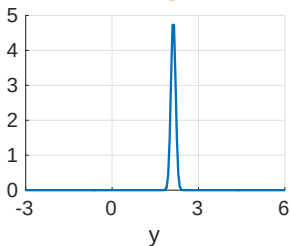


EXAMPLE 2 – WHAT IS $p(y)$ =?

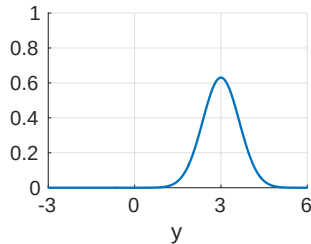
CHALMERS



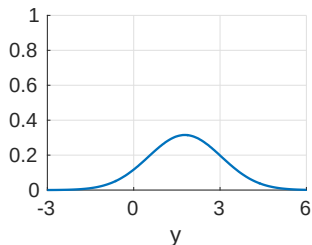
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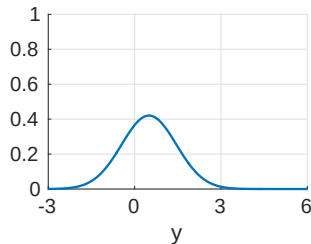
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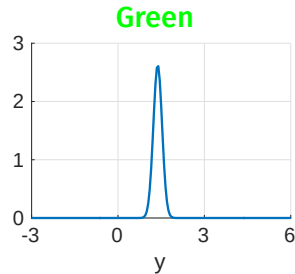
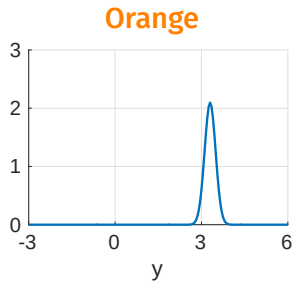
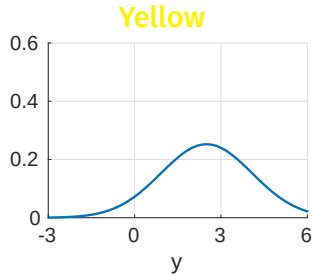
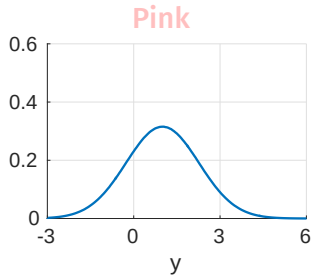
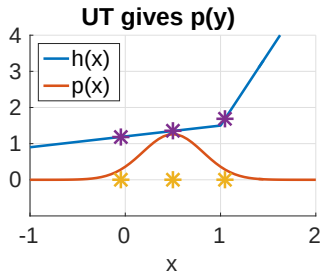


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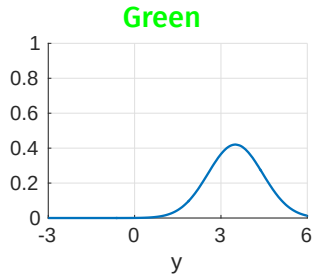
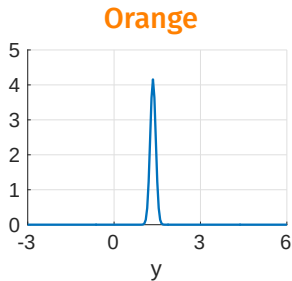
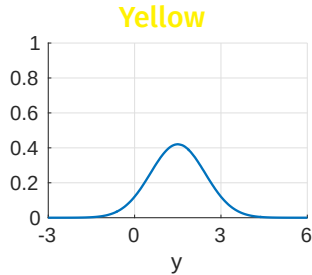
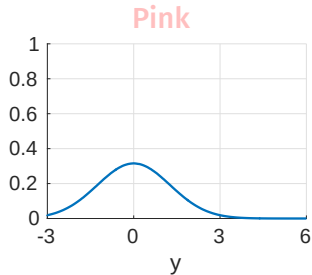
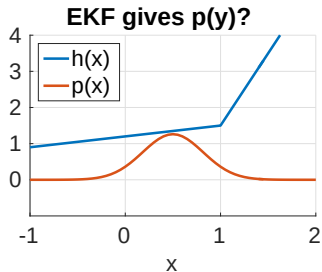
EXAMPLE 3 – WHAT IS $p(y)$ =?

CHALMERS



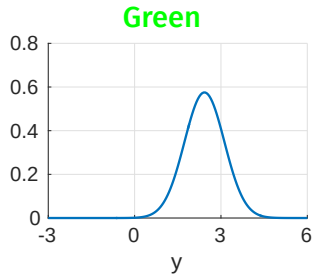
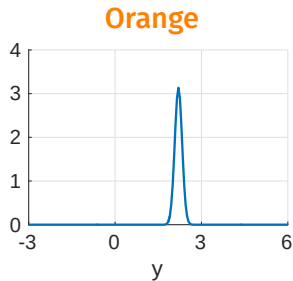
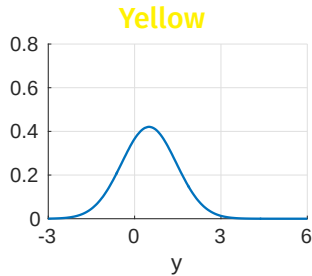
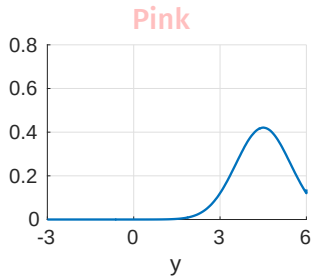
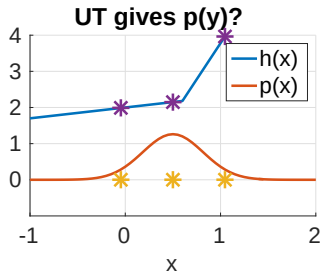
EXAMPLE 4 – WHAT IS $p(y)$ =?

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EXAMPLE 5 – WHAT IS $p(y)$ =?

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EXAMPLE 6 – WHAT IS $p(y)$ =?

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